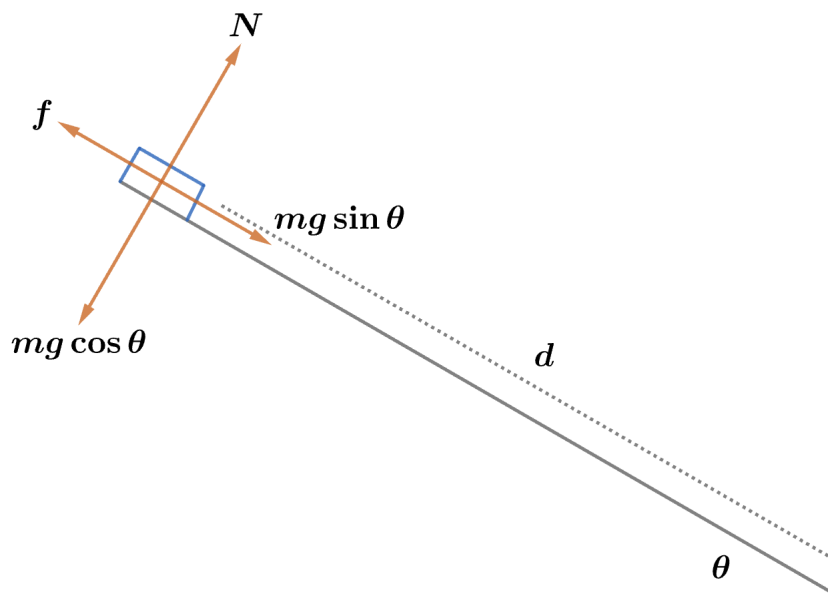


2007 F=ma Exam: Problem 26

Kevin S. Huang



Applying Newton's 2nd law to the sled,

$$mg \sin \theta - f = ma$$

Since $f = \mu N = \mu mg \cos \theta$,

$$mg \sin \theta - \mu mg \cos \theta = ma$$

$$a = g(\sin \theta - \mu \cos \theta)$$

If the sled travels at acceleration a for time t starting from rest, the distance it covers is

$$d = \frac{1}{2}at^2 = \frac{gt^2}{2}(\sin \theta - \mu \cos \theta)$$

Solving for μ ,

$$\frac{2d}{gt^2} = \sin \theta - \mu \cos \theta$$

$$\mu = \frac{1}{\cos \theta} \left(\sin \theta - \frac{2d}{gt^2} \right) = \frac{1}{\cos 25^\circ} \left(\sin 25^\circ - \frac{2(85 \text{ m})}{(10 \text{ m/s}^2)(17 \text{ s})^2} \right) = 0.40$$

so the answer is B.